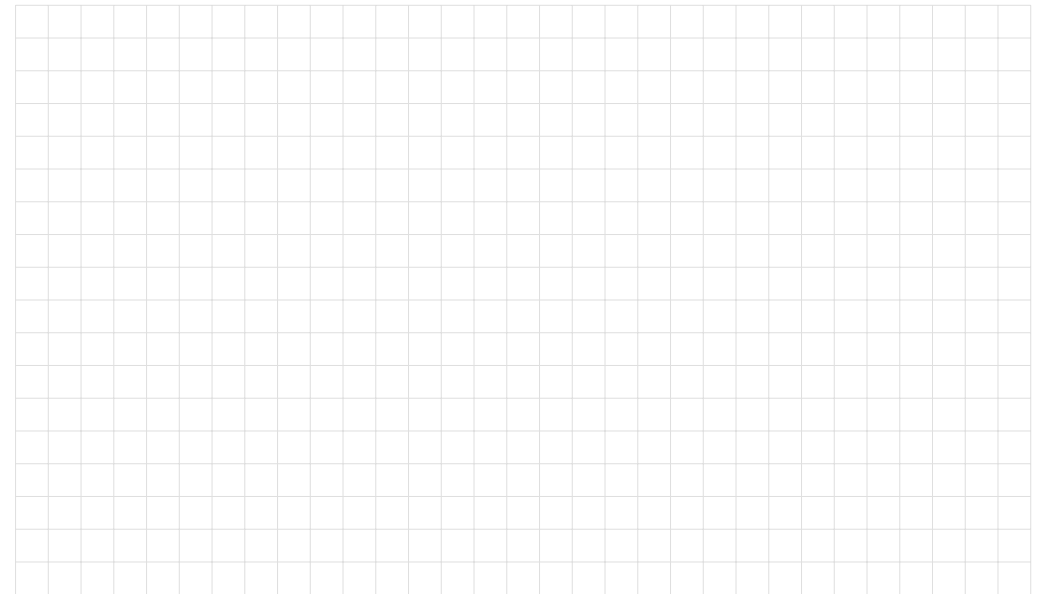
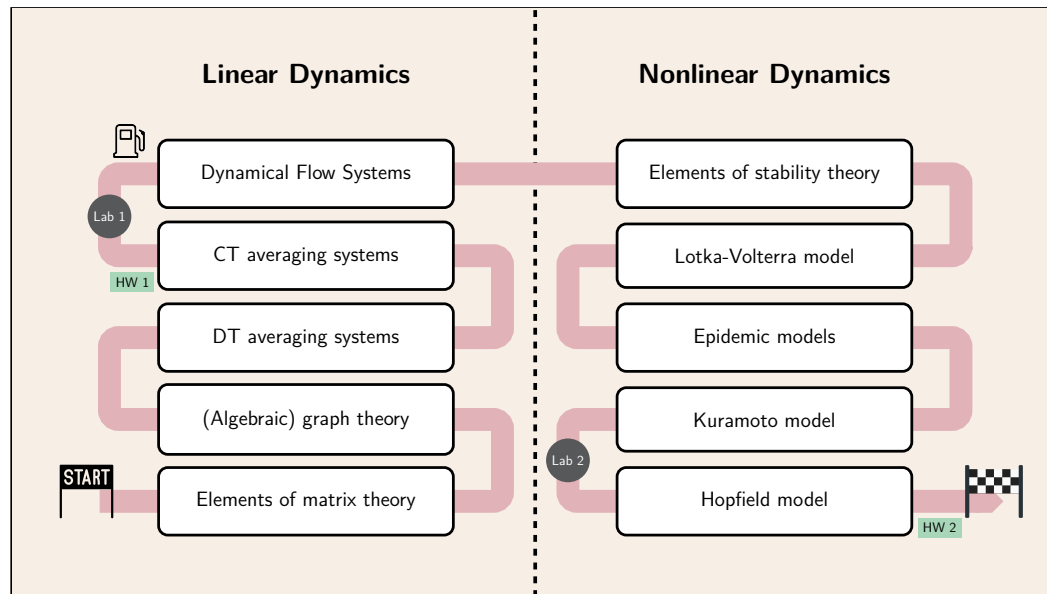
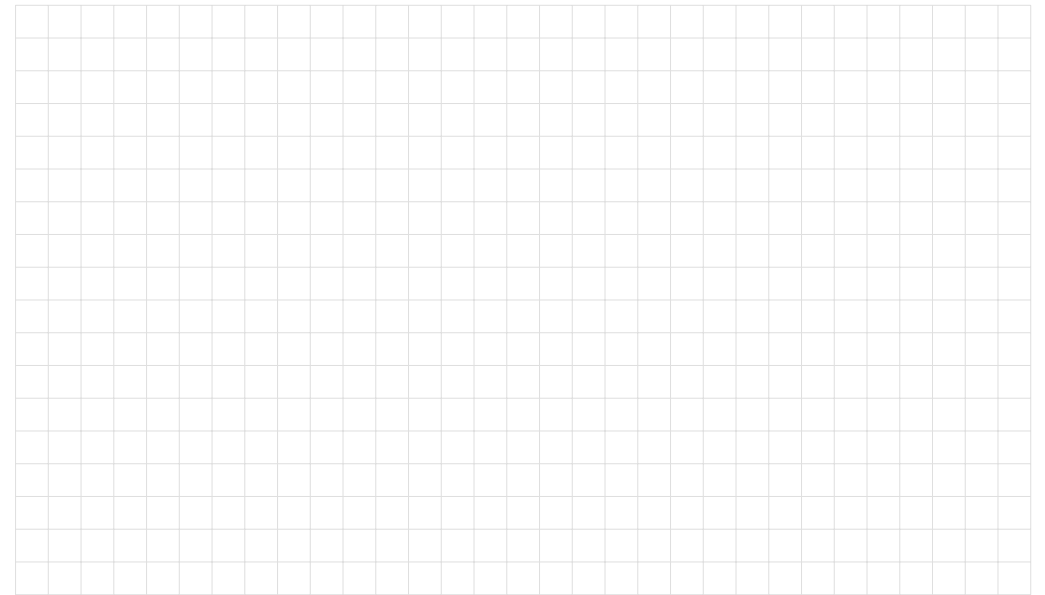


Elements of Algebraic Graph Theory

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Network Dynamical Systems
Master Degree in Control Systems Engineering
A.A. 2025-2026



Covered topics

- ▷ DT averaging systems: recap and examples
- ▷ Preliminaries on substochastic matrices
- ▷ Systems achieving asymptotic consensus
- ▷ Systems reaching asymptotic disagreement



DT averaging systems

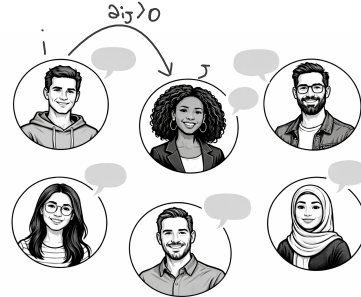
$$x(k+1) = Ax(k), \quad x(0) = x_0 \in \mathbb{R}^n$$

- ▷ A adjacency matrix of weighted digraph $G = (V, E, \{a_{ij}\})$
- ▷ A row-stochastic

$$\lim_{k \rightarrow \infty} x(k) = \lim_{k \rightarrow \infty} A^k x(0) \quad \begin{cases} \text{does not exist} & \Rightarrow \text{not converging} \\ = \alpha \mathbf{1}_n, \alpha \in \mathbb{R} & \Rightarrow \text{asymptotic consensus} \\ = v \neq \alpha \mathbf{1}_n, \alpha \in \mathbb{R} & \Rightarrow \text{asymptotic disagreement} \end{cases}$$

DT averaging systems: social networks

- ▷ individuals $V = \{1, 2, \dots, n\}$
- ▷ $(i, j) \in E$ if i is influenced by j
- ▷ $x_i \in \mathbb{R}$ = opinion that individual i has on a certain topic



French-Harary-DeGroot model

Opinion of i is updated as a weighted average of neighbors' opinions:

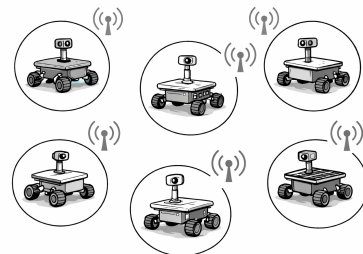
$$x_i(k+1) = \sum_{j=1}^n a_{ij} x_j(k), \quad \sum_{j=1}^n a_{ij} = 1$$

$$\iff x(k+1) = Ax(k), \quad A \text{ row-stochastic}$$

Q: do all individuals asymptotically agree on the same opinion?

DT averaging systems: robotic networks

- ▷ mobile robots $V = \{1, 2, \dots, n\}$
- ▷ $(i, j) \in E$ if i receives information from j
- ▷ $x_i \in \mathbb{R} = (1-d)$ position of i



Discrete-time rendezvous model

Robot i moves toward the positions of its neighbors according to:

$$x_i(k+1) = x_i(k) + \sum_{j=1}^n a_{ij} \overbrace{(x_j(k) - x_i(k))}^{\text{relative displacement}}, \quad \sum_{j=1}^n a_{ij} = 1$$

$$\iff x(k+1) = Ax(k), \quad A \text{ row-stochastic}$$

Q: do all robots asymptotically meet at the same position?

$$x_i(k+1) = x_i(k) + \sum_{j=1}^n a_{ij} x_j(k) - \underbrace{\sum_{j=1}^n a_{ij}}_{=1} x_i(k) = \sum_{j=1}^n a_{ij} x_j(k)$$

DT averaging systems: the story so far

A row-stochastic ($D_{\text{out}} = I$)

A primitive G strongly connected and aperiodic	A irreducible but not primitive G strongly connected and periodic
* 1 is a simple, strictly dominant eigenvalue of A but A not primitive $G ??$	??? Asymptotic disagreement/not converging

↓
asymptotic consensus

* ie: $A = \begin{bmatrix} 0.5 & 0.5 \\ 0 & 1 \end{bmatrix}$

Preliminaries on substochastic matrices

Def: A non-negative matrix $A \in \mathbb{R}^{n \times n}$ is row-substochastic if its row-sums are at most 1 and at least one row-sum is strictly less than 1. That is:

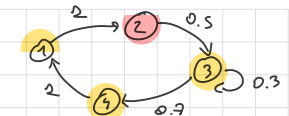
$$A\mathbf{1}_n \leq \mathbf{1}_n \quad \text{and} \quad \exists i \in \{1, \dots, n\} \text{ s.t. } e_i^\top A\mathbf{1}_n < 1.$$

Lemma (Convergent row-substochastic matrices): Let G be a weighted digraph with row-substochastic adjacency matrix A . Then, A is convergent if and only if G contains directed walks from each node with (weighted) out-degree 1 to a node with (weighted) out-degree strictly less than 1.

$\rightarrow \lim_k A^k = 0$

▷ Corollary: A row-substochastic + G strongly connected $\implies A$ convergent

Consider row-stochastic matrix $A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0.5 & 0 \\ 0 & 0 & 0.3 & 0.7 \\ 2 & 0 & 0 & 0 \end{bmatrix}$



● nodes with out-degree 1

● nodes with out-degree < 2

Lemma (Convergent row-substochastic matrices): Let G be a weighted digraph with row-substochastic adjacency matrix A . Then, A is convergent if and only if G contains directed walks from each node with (weighted) out-degree 1 to a node with (weighted) out-degree strictly less than 1.

Proof:

" G contains walks from each node with out-degree 1 to a node with out-degree $< 1 \Rightarrow A$ convergent"

Define $S_h := \{i \in V \mid e_i^T A^h \mathbf{1}_n \leq 1\}$

We prove the following facts: ① $S_h \subseteq S_{h+2} \quad \forall h$

② i has an out-neighbor $j \in S_h \Rightarrow i \in S_{h+1}$

③ $\exists \bar{h}$ s.t. $S_{\bar{h}} = V$

① Let $i \in S_h$, then $e_i^T A^{h+1} \mathbf{1} = e_i^T A^h A \mathbf{1}_n \leq e_i^T A^h \mathbf{1}_n < 1$ $\because i \in S_h$
 ≤ 1 since A is row substochastic

This implies that $i \in S_{h+1}$

② Note that (i) $A \mathbf{1} \leq \mathbf{1}$

(ii) $j \in S_h \Rightarrow e_j^T A^h \mathbf{1} < 1$
 $\hookrightarrow 1 - (1 - e_j^T A^h \mathbf{1}) > 0$

Therefore, if $j \in S_h$ $e_i^T A^{h+1} \mathbf{1} = (e_i^T A)(A^h \mathbf{1}) \stackrel{(i)}{\leq} (e_i^T A)(\mathbf{1} - (1 - e_j^T A^h \mathbf{1})e_j) \equiv \underbrace{e_i^T A \mathbf{1}}_{\leq 1} - \underbrace{(1 - e_j^T A^h \mathbf{1})}_{> 0} \underbrace{e_i^T A e_j}_{> 0}$ $< 1 \Rightarrow i \in S_{h+1}$
 $\stackrel{(ii)}{\Rightarrow} > 0$ Since by assumption, i has an out-neighbor $j \in S_h$

③ Note that, by definition, S_1 contains the nodes with out-degree ≤ 1 . By facts ① & ②, S_2 contains the nodes in S_1 and the nodes in $V \setminus S_1$ that have out-neighbor in S_1

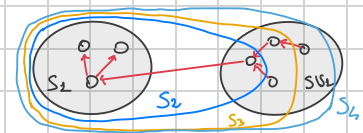
Similarly, S_3 contains the nodes in S_2 and the nodes in $V \setminus S_2$ that have an out-neighbor in S_2 , that is the nodes in $V \setminus S_2$ that are connected by a walk of length 2 to nodes in S_1 . By iterating

this argument, since by assumption $\forall$ nodes $V \setminus S_1 \exists$ walk to a node in $S_1 \Rightarrow \exists \bar{h}$ s.t. $S_{\bar{h}} = V$

To conclude, pick $\bar{h}$ that satisfies $S_{\bar{h}} = V$ and define $\gamma = \max_{i \in \{1, \dots, n\}} \sum_{j=1}^{\bar{h}} [A^{\bar{h}}]_{ij} < 1$

For every $K \geq \bar{h}$, we can write $K = q\bar{h} + r$, where " q " is quotient and " $r \in \{0, \dots, \bar{h}-1\}$ " remainder $A^K \mathbf{1} = A^{q\bar{h}+r} \mathbf{1} \leq A^{q\bar{h}} \mathbf{1} \leq \gamma^q \mathbf{1} \xrightarrow[q \rightarrow \infty]{K \rightarrow \infty} 0$

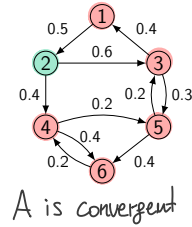
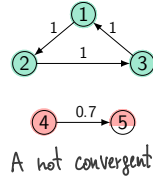
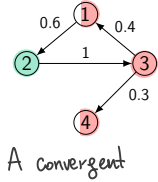
Thus, as $K \rightarrow \infty$, we have that $A^K \mathbf{1} \rightarrow 0$ and since $A \geq 0$ it follows that $\lim_{K \rightarrow \infty} A^K = 0$



Preliminaries on substochastic matrices

Q: Is the adjacency matrix of the following weighted digraphs convergent?

● = nodes with out-degree = 1
 ● = nodes with out-degree < 2



DT averaging systems achieving asymptotic consensus

Thm (Consensus in DT averaging systems): Let G be a weighted digraph with row-stochastic adjacency matrix A . The following statements are equivalent:

- (A1) the eigenvalue 1 is simple and all other eigenvalues μ satisfy $|\mu| < 1$;
- (A2) A is semi-convergent and $\lim_{k \rightarrow \infty} A^k = \mathbb{1}_n w^\top$ where $w \in \mathbb{R}^n$ satisfies $w \geq 0$, $\mathbb{1}_n^\top w = 1$, and $w^\top A = w^\top$;
- (A3) G contains a globally reachable node and the subgraph of globally reachable nodes is aperiodic. *If there is a self loop $\Rightarrow G$ aperiodic*

If any, and therefore all, of the previous conditions are satisfied, then

- (i) w is the left dominant eigenvector of A and $w_i > 0$ iff node i is globally reachable;
- (ii) the solution of the DT averaging system satisfies $\lim_{k \rightarrow \infty} x(k) = (w^\top x(0)) \mathbb{1}_n$.

▷ Corollary: A doubly-stochastic + (A3) $\Rightarrow w = \frac{1}{n} \mathbb{1}_n$, $\lim_{k \rightarrow \infty} x(k) = \frac{1}{n} \sum_{i=1}^n x_i(0) \mathbb{1}_n$
average of init condition

Notes: 1) The equivalence $(A_1) \Leftrightarrow (A_2)$ has been already proved (last thm of module 1)

2) Set of globally reachable nodes is strongly connected

3) In general the consensus value $w^\top x(0)$ doesn't depend on init. cond of all nodes, but only on the initial conditions of globally reachable nodes

Proof: (A2) $\Leftrightarrow$ (A3)

"(A2) $\Rightarrow$ (A3)"

Since $\mathbb{1}^T W = 1$, then at least one component of w must be strictly positive, say component w_j

Since $\lim_{K \rightarrow \infty} A^K = \mathbb{1} w^T$, this means that the j -th column of $\lim_{K \rightarrow \infty} A^K$ must be a positive vector $\Rightarrow$ for sufficiently large K we have that the j -th column of A^K is a positive vector

Equivalently, there is a walk of length K from every node to node j . Hence node j is globally reachable.

Moreover, we have that $[A^K]_{ij}$ will converge to a positive number as $K \rightarrow \infty$. Then there are walks of length K and $K+1$ for a sufficiently large K . This implies that the period of j must be equal to 1.

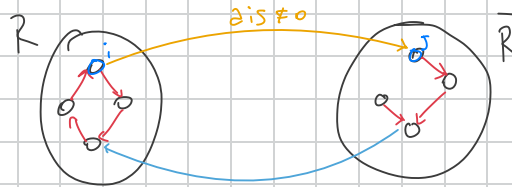
Consider now the subgraph of globally reachable nodes, which is strongly connected. It can be shown that in a strongly connected (sub)graph all nodes have same period.

Hence the period of all globally reachable nodes is 1 and therefore, by a previous lemma, the subgraph of globally reachable nodes is aperiodic.

"(A3) $\Rightarrow$ (A2)"

Let R be the set of globally reachable nodes and $\bar{R} = V \setminus R$. By assumption, $R \neq \emptyset$

Moreover, for every $i \in R$ and $j \in \bar{R}$, it holds that $a_{ij} = 0$ otherwise j would be globally reachable



Up to a re-labeling of nodes: $A = \begin{bmatrix} A_{11} & 0 \\ A_{12} & A_{22} \end{bmatrix}$ where $n_1 = |R|$ and $n_2 = |\bar{R}|$

where A_{11} is the adj. matrix of subgraph of globally reachable nodes, G_R ,

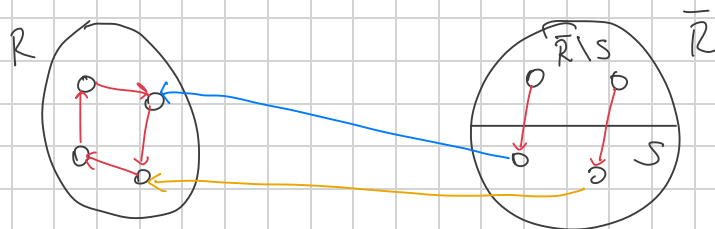
and A_{22} is that of the subgraph of the nodes in $\bar{R}$, $G_{\bar{R}}$.

Since G_R is strongly connected and, by assumption, aperiodic, then A_{11} is primitive and $\lim_{K \rightarrow \infty} A_{11}^K = \mathbb{1}_n w_n^T$ where $w_n \geq 0$ is the left dominant eigenvector of A_{11}

such that $\mathbb{1}_n^T w_n = 1$.

Since $A_{22} \neq 0$, A_{22} is a row-substochastic matrix. Define $S = \{i \in \bar{R} : \exists j \in R \text{ with } a_{ij} > 0\} = \{i \in \bar{R} : \sum_{j \in R} a_{ij} > 0\} \rightarrow \begin{bmatrix} A_{11} & 0 \\ a_{15} & \text{sum} < 2 \end{bmatrix}$

Pick any $i \in \bar{R}$ then $\exists$ a walk from i to $j \in R$. But then this walk must visit a node in S



So, by the Lemma on row-stochastic matrices, A_{22} is convergent.

Since A is block triangular, $\text{spec}(A) = \text{spec}(A_{11}) \cup \text{spec}(A_{22})$, so A has a simple and strictly dominant eigenvalue in 1

Then A is semi-convergent: $A^K = \begin{bmatrix} A_{11}^K & 0 \\ * & A_{22}^K \end{bmatrix} \rightarrow A^\infty = \begin{bmatrix} A_{11}^\infty & 0 \\ A_{21}^\infty & A_{22}^\infty \end{bmatrix}$ finite

where $A_{11}^\infty = \mathbb{1}_{n_1} w_1^T$ and $A_{22}^\infty = 0$. It remains to compute A_{21}^∞ .

Note that $A^{K+1} \xrightarrow{K \rightarrow \infty} A^\infty$ $A \cdot A^K \xrightarrow{K \rightarrow \infty} A \cdot A^\infty$

Thus: $A^\infty = A \cdot A^\infty$ and

$$\begin{bmatrix} \overbrace{A_{11}^\infty}^{\mathbb{1}_{n_1} w_1^T} & 0 \\ A_{21}^\infty & \underbrace{A_{22}^\infty}_0 \end{bmatrix} = \begin{bmatrix} A_{11} & 0 \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} \overbrace{A_{11}^\infty}^{\mathbb{1}_{n_1} w_1^T} & 0 \\ A_{21}^\infty & \underbrace{A_{22}^\infty}_0 \end{bmatrix}$$

$$\Rightarrow A_{21}^\infty = A_{21} \mathbb{1}_{n_1} w_1^T + A_{22} A_{21}^\infty \Rightarrow (I - A_{22}) A_{21}^\infty = A_{21} \mathbb{1}_{n_1} w_1^T$$

$$\xRightarrow{p(A_{22}) < 1} A_{21}^\infty = (I - A_{22})^{-1} A_{21} \mathbb{1}_{n_1} w_1^T \quad (*)$$

Note that since A row-stochastic: $A \mathbb{1}_n = \mathbb{1}_n \Rightarrow \begin{bmatrix} A_{11} & 0 \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} \mathbb{1}_{n_1} \\ \mathbb{1}_{n_2} \end{bmatrix} = \begin{bmatrix} \mathbb{1}_{n_1} \\ \mathbb{1}_{n_2} \end{bmatrix} \Rightarrow A_{21} \mathbb{1}_{n_1} + A_{22} \mathbb{1}_{n_2} = \mathbb{1}_{n_2}$
 $\Rightarrow A_{21} \mathbb{1}_{n_1} = (I - A_{22}) \mathbb{1}_{n_2}$
 $\Rightarrow (I - A_{22})^{-1} A_{21} \mathbb{1}_{n_1} = \mathbb{1}_{n_2}$

By plugging last identity in $(*)$: $A_{21}^\infty = \mathbb{1}_{n_2} w_1^T$

To conclude, we have $\lim_{K \rightarrow \infty} A^K = \begin{bmatrix} A_{11}^\infty & 0 \\ A_{21}^\infty & A_{22}^\infty \end{bmatrix} = \begin{bmatrix} \mathbb{1}_{n_1} w_1^T & 0 \\ \mathbb{1}_{n_2} w_1^T & 0 \end{bmatrix} = \mathbb{1}_n \begin{bmatrix} w_1 \\ 0 \end{bmatrix}^T$ where $w \geq 0$ s.t. $\mathbb{1}_n^T w = \mathbb{1}_{n_1}^T w_1 = 1$

Finally, a direct computation shows that $w^T A = w^T$, that is w is the left dominant eigenvector of A , and from above proof it follows that $w_i > 0$: is globally reach.

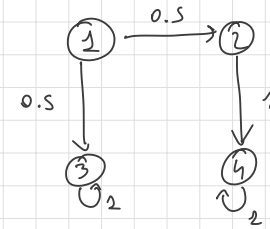
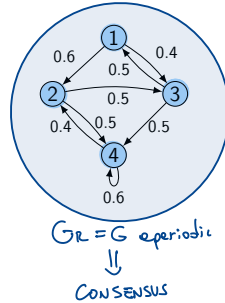
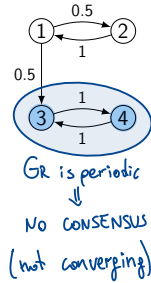
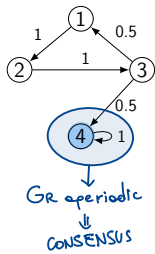
So this proves statement (i)

To prove statement (ii) note that: $x(k) = A^k x(0) \xrightarrow{k \rightarrow \infty} A^\infty x(0) = \underbrace{\mathbb{1}_n^T W^T x(0)}_{\text{scalar}} = (w^T x(0)) \mathbb{1}_n$

DT averaging systems achieving asymptotic consensus

Q: Do the systems described by the following digraphs achieve asymptotic consensus?

● = globally reach nodes



Not globally reachable node $\Rightarrow$ NO CONSENSUS

$$A = \begin{bmatrix} 0 & 0.5 & 0.5 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

$\rightarrow$ eigenvalues 0, 2

2 semi-simple $\Rightarrow A$ semi-convergent
 $\Rightarrow A^k \xrightarrow{k \rightarrow \infty} A^\infty$ finite

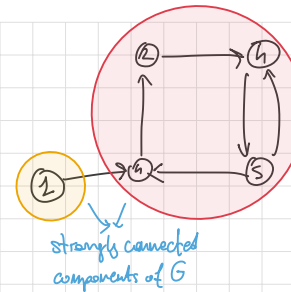
Condensation digraph

Def: Given a (weighted) digraph G . A subgraph S is a strongly connected component of G if S is strongly connected and any other subgraph of G strictly containing S is not strongly connected.

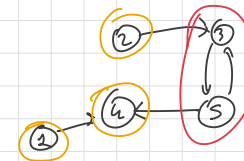
Def: The **condensation digraph** $C(G)$ of a (weighted) digraph G is obtained by collapsing each strongly connected component of G into a single node. For any two distinct strongly connected components S_k and S_h there is an edge (S_k, S_h) in $C(G)$ iff there exists an edge (i, j) in G with $i \in S_k$ and $j \in S_h$.

▷ **Note:** If G contains a globally reachable node then $C(G)$ has a unique sink

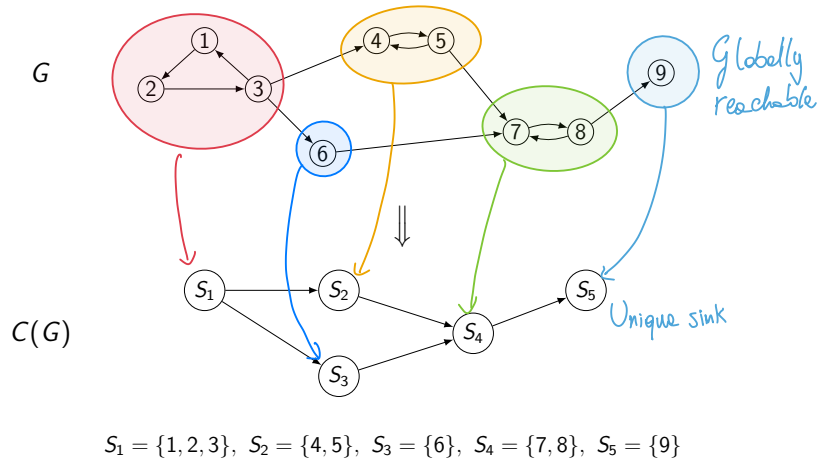
Self loop not allowed in $C(G)$



Note: A single node (with or without self loop) is strongly connected



Condensation digraph



DT averaging systems reaching asymptotic disagreement

Thm (Disagreement in DT averaging systems): Let G be a weighted digraph with row-stochastic adjacency matrix A . Assume that the condensation digraph $C(G)$ has $n_s \geq 2$ sinks. The following statements are equivalent:

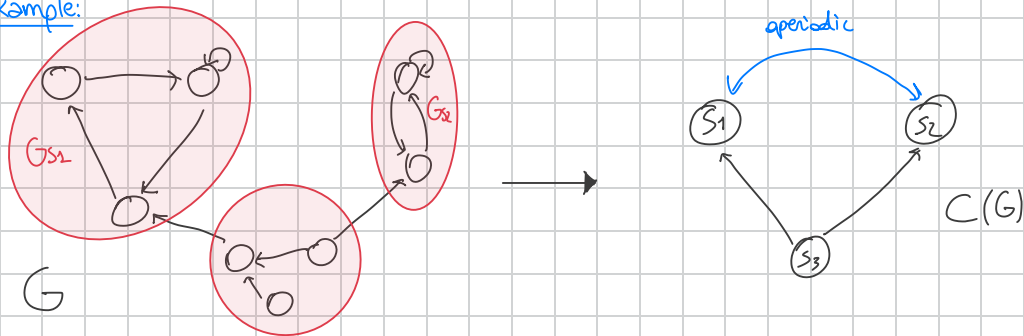
- (A1) the eigenvalue 1 is semi-simple with multiplicity n_s and all other eigenvalues μ satisfy $|\mu| < 1$;
- (A2) A is semi-convergent;
- (A3) each sink of $C(G)$ is aperiodic.

If any, and therefore all, of the previous conditions are satisfied, then the solution of the DT averaging system satisfies

$$\lim_{k \rightarrow \infty} x(k) = v \neq \alpha \mathbf{1}_n, \alpha \in \mathbb{R}.$$

disagreement

Example:



adj matrix

$$A = \begin{bmatrix} A_{11} & 0 & 0 \\ 0 & A_{22} & 0 \\ A_{31} & A_{32} & A_{33} \end{bmatrix}$$

- with
- A_{11}, A_{22} row stoch be the adj matrices of G_{s1}, G_{s2}
 - A_{33} row substoch and the adj matrix of G_{s3}
 - A_{11}, A_{22} prim, A_{33} convergent

$$A_{11}^K \xrightarrow{K \rightarrow \infty} \mathbb{1}_{n_1} w_1^T$$

$$A_{22}^K \xrightarrow{K \rightarrow \infty} \mathbb{1}_{n_2} w_2^T$$

$$A_{33}^K \xrightarrow{K \rightarrow \infty} 0$$

$w_1, w_2 \geq 0$ left dominant eigenvectors of A_{11}, A_{22}

It can be shown that $A_{32}^K \rightarrow (I - A_{33})^{-2} A_{32} \mathbb{1}_{n_2} w_2^T$

$$A_{31}^K \rightarrow (I - A_{33})^{-2} A_{31} \mathbb{1}_{n_1} w_1^T$$

$$\text{with } (I - A_{33})^{-2} A_{31} \mathbb{1}_{n_1} + (I - A_{33})^{-2} A_{32} \mathbb{1}_{n_2} = \mathbb{1}_{n_3}$$

Thus: $x(K) = A^K x(0) \xrightarrow{K \rightarrow \infty} \begin{bmatrix} \mathbb{1}_{n_1} w_1^T & 0 & 0 \\ 0 & \mathbb{1}_{n_2} w_2^T & 0 \\ A_{31}^\infty & A_{32}^\infty & 0 \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_2(0) \\ x_3(0) \end{bmatrix} = \begin{bmatrix} \mathbb{1}_{n_1} w_1^T x_1(0) \\ \mathbb{1}_{n_2} w_2^T x_2(0) \\ A_{31}^\infty x_1(0) + A_{32}^\infty x_2(0) \end{bmatrix} = \begin{bmatrix} w_1^T x_1(0) \mathbb{1}_{n_1} \\ w_2^T x_2(0) \mathbb{1}_{n_2} \\ \boxed{w_2^T x_2(0) \mathbb{1}_{n_2}} \end{bmatrix}$

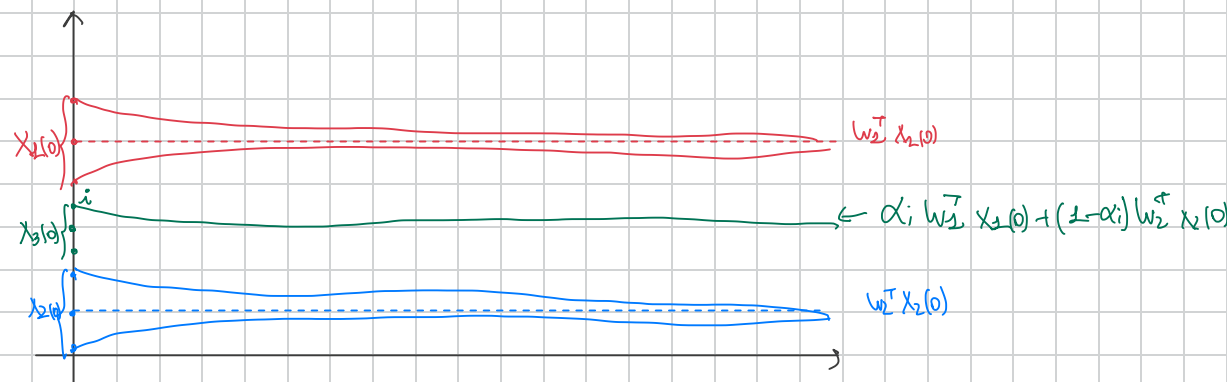
$\begin{bmatrix} \mathbb{1}_{n_1} w_1^T x_1(0) \\ \mathbb{1}_{n_2} w_2^T x_2(0) \\ A_{31}^\infty x_1(0) + A_{32}^\infty x_2(0) \end{bmatrix}$

$A_{31}^\infty x_1(0) + A_{32}^\infty x_2(0) = (I - A_{33})^{-2} A_{31} \mathbb{1}_{n_1} w_1^T x_1(0) + (I - A_{33})^{-2} \mathbb{1}_{n_2} w_2^T x_2(0)$

$= w_1^T x_1(0) (I - A_{33})^{-2} A_{31} \mathbb{1}_{n_1} + w_2^T x_2(0) (I - A_{33})^{-2} A_{32} \mathbb{1}_{n_2}$

$[x_3(\infty)]_i = \alpha_i w_{2i}^T x_2(0) + (1 - \alpha_i) w_{2i}^T x_2(0) \quad 0 \leq \alpha_i \leq 1$

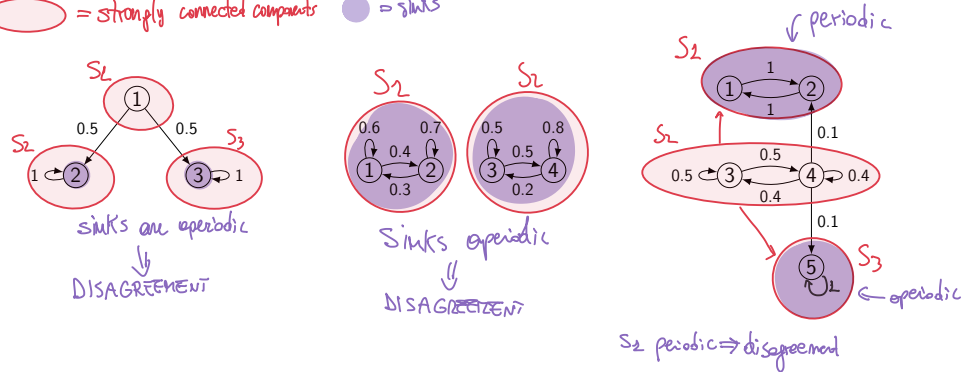
Convex comb. of consensus value of G_{s1} & G_{s2}



DT averaging systems reaching asymptotic disagreement

Q: Do the systems described by the following digraphs reach asymptotic disagreement?

○ = strongly connected components ● = sinks



DT averaging systems: Rosetta stone

A row-stochastic ($D_{\text{out}} = I$)

① A primitive G strongly connected and aperiodic	② A irreducible but not primitive G strongly connected and periodic
③ 1 is a simple, strictly dominant eigenvalue of A but A not primitive C(G) has 1 aperiodic sink	④ 1 is a semi-simple but not simple, strictly dominant eigenvalue of A C(G) has ≥ 2 aperiodic sinks
	⑤

G has at least one globally reachable node and the subgraph of globally reachable nodes is aperiodic

asymptotic consensus

asymptotic disagreement

not converging

$$\textcircled{1} \quad A = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$$

$$\textcircled{2} \quad A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\textcircled{3} \quad A = \begin{bmatrix} 2 & 0 \\ 1/2 & 1/2 \end{bmatrix}$$

$$\textcircled{4} \quad A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\textcircled{5} \quad A = \left[\begin{array}{cc|c} 0 & 1 & 0 \\ 1 & 0 & 0 \\ \hline 0 & 0 & 1 \end{array} \right]$$